

newer versions of the Windows OS, and there were issues with compatibility.

A software development committee created jointly by DCS and NIC was against the hidden costs of proprietary software. They evaluated Lypsaa and submitted a proposal to migrate it to an open source solution named OpenLypsaa. The committee started its efforts to develop free software using hired resources in 2007.

Open source for cost-cutting

The most prominent benefit to the societies is related to cost. “As a result of using free and open source software at every possible step, the Kerala DCSs collectively saved about Rs 25 lakh,” states Simon E K, District Informatics Officer, NIC. “If we went ahead with Windows-based PCs and proprietary databases, it would have cost about Rs 5000 per society. We are assured that our total investment of Rs 5-6 lakh was a good one. We also have a hold on the total cost of ownership including development, installing and using software for the duration of its life,” he adds. The DCSs did not have to make additional investments for new computer systems as they could use their regular systems with Intel architecture, 512 MB or higher RAM, and 80 GB hard disk drives.

Ease of maintenance is another advantage that Jose Emmanuel, deputy director, Dairy Development, Government of Kerala, noted. He states, “With Microsoft, we would often get calls from the societies due to viruses on their machines—at least one call per month. Our experience with open source proves that it is much more secure, as we receive no complaints about viruses, reformatting and system crashes.”

Going the open source way

To begin with, Ubuntu was introduced on all the DCS systems. OpenLypsaa was developed and implemented at 30 societies by 2008. It is a complete software solution developed using the LAMP platform. OpenLypsaa is used in functions like generating milk-bills for farmers and taking care of their daily accounting. It is used to register members, keep milk procurement details, manage financial accounting, prepare milk bills, register any service connections, and maintain the accounts and details of various customers. “Milk bills include details like the quantity of produce brought, its quality, the price at which it was sold, etc. When data was entered manually, it was impossible to avoid numerical errors. Mistakes are almost negligible now. Moreover, there was room for malpractice when it came to creating a voters’ list for elections within the societies. This system ensures fair play by filtering out farmers who have not contributed a minimum amount of milk to the society,” Khalid confidently states.

The system maintains registers for sales, producers’ personal details, cattle feed sales, advance payments and financial accounts, and also helps in generating crucial reports on voters, pensioners and scheme beneficiaries. “For instance, the state government sometimes gives benefits to farmers who have contributed regularly for two months, or have given extra produce. It was earlier a cumbersome task to look through our



The screenshot shows the 'Profile' page of the OpenLypsaa application. It includes a form for user details and a table listing members.

S.No	Name	Age	Gender	Address	Phone No.	Registration No.	Status
1	K. S. S. S.	25	M	...	...	...	...
2	K. S. S. S.	25	M	...	...	...	...
3	K. S. S. S.	25	M	...	...	...	...
4	K. S. S. S.	25	M	...	...	...	...
5	K. S. S. S.	25	M	...	...	...	...
6	K. S. S. S.	25	M	...	...	...	...
7	K. S. S. S.	25	M	...	...	...	...
8	K. S. S. S.	25	M	...	...	...	...
9	K. S. S. S.	25	M	...	...	...	...
10	K. S. S. S.	25	M	...	...	...	...
11	K. S. S. S.	25	M	...	...	...	...
12	K. S. S. S.	25	M	...	...	...	...
13	K. S. S. S.	25	M	...	...	...	...
14	K. S. S. S.	25	M	...	...	...	...
15	K. S. S. S.	25	M	...	...	...	...
16	K. S. S. S.	25	M	...	...	...	...
17	K. S. S. S.	25	M	...	...	...	...
18	K. S. S. S.	25	M	...	...	...	...
19	K. S. S. S.	25	M	...	...	...	...
20	K. S. S. S.	25	M	...	...	...	...
21	K. S. S. S.	25	M	...	...	...	...
22	K. S. S. S.	25	M	...	...	...	...
23	K. S. S. S.	25	M	...	...	...	...
24	K. S. S. S.	25	M	...	...	...	...
25	K. S. S. S.	25	M	...	...	...	...
26	K. S. S. S.	25	M	...	...	...	...
27	K. S. S. S.	25	M	...	...	...	...
28	K. S. S. S.	25	M	...	...	...	...
29	K. S. S. S.	25	M	...	...	...	...
30	K. S. S. S.	25	M	...	...	...	...



OpenLypsaa is a software solution developed using the LAMP platform, which is being used by the Kerala Dairy Cooperative Societies for functions like registering members, generating milk-bills for farmers, keeping milk procurement details, managing financial accounting, and so on

registers and fish through all the entries. With OpenLypsaa, the beneficiaries are a click away. We can also generate a periodical report on farmers based on these entries,” adds Khalid.

OpenLypsaa used TCPDF—a PHP class to generate PDF documents, ‘Business Intelligence and Reporting Tool’ (BIRT) for report generation, and HighCharts for charting MIS reports. HighCharts is a charting library written in pure JavaScript, which offers an easy way to add interactive charts to a website or Web application. Khalid says, “All data and configuration comes in JSON format. It fits well with PHP, without having to write tons of code or use some third-party library. The `json_encode` function is very handy when passing options from PHP to HighCharts.”

The committee, along with NIC, also developed a solution called ‘System for Automating Milk Procurement’ (SAMP). It was